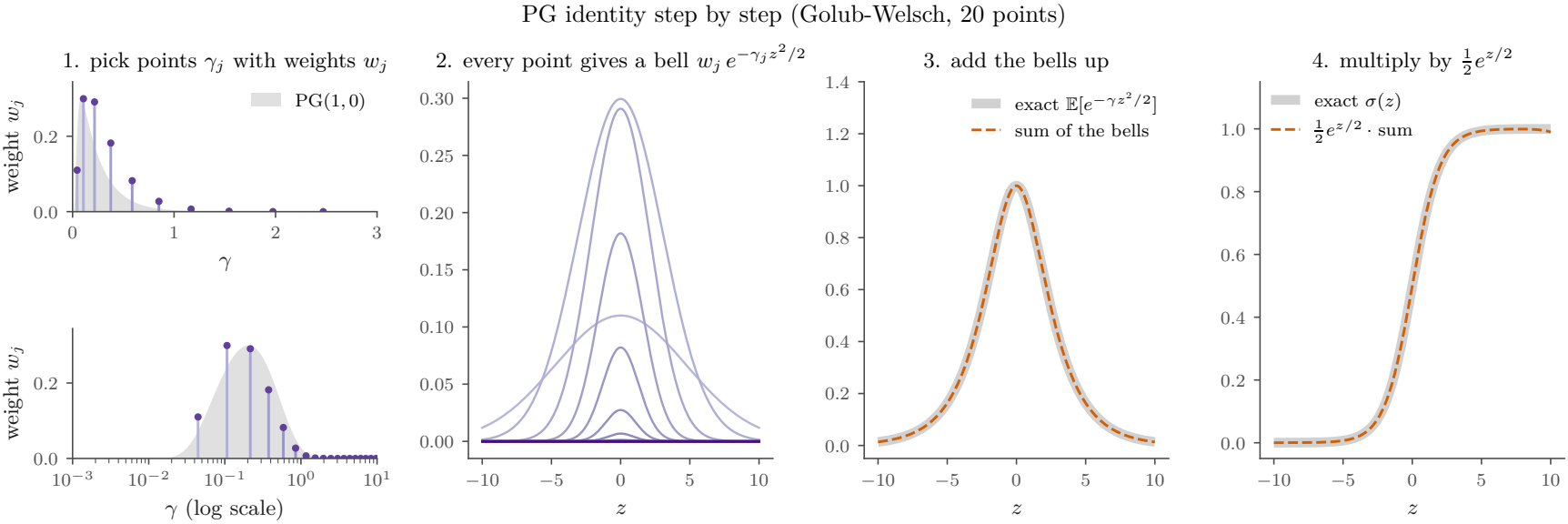

PG identity step by step (Golub-Welsch, 20 points)
1. pick points γ_j with weights w_j
PG(1, 0)
weight w_j
γ
γ (log scale)
2. every point gives a bell $w_j\, e^{-\gamma_j z^2/2}$
z
3. add the bells up
exact $\mathbb{E}[e^{-\gamma z^2/2}]$
sum of the bells
z
4. multiply by $\frac{1}{2}e^{z/2}$
exact $\sigma(z)$
$\frac{1}{2}e^{z/2} \cdot$ sum
z